



BACHELOR OF INFORMATION TECHNOLOGY (HONS)

RESIT EXAMINATION
MAY 2019

Course : EC3114 (Software Modeling and Analysis)

Time : 1:00pm – 4:00pm
(3 hours)

Lecturer : Wan Nor Al-Ashekin Binti Wan Husin

Date : 15 December 2019

Instructions:

Answer **ALL** questions in the Answer Booklet provided.

The maximum number of marks is 100.

This question paper consists of 2 printed pages
excluding front cover).

Cheating is a very serious offence and those caught will be severely punished

DO NOT OPEN THIS BOOKLET UNTIL YOU ARE TOLD TO DO SO

SECTION A

[50 MARKS]

1. List FIVE (5) methods for fact finding in requirements engineering. (5 marks)
2. Explain TWO (2) System Analyst's tasks during project initiation and analysis stage. (4 marks)
3. Explain THREE (3) different types of stakeholders who might be interested in new system development and could help in providing requirements. (6 marks)
4. Give any TWO (2) examples of the following non-functional requirements categories:
 - Non-functional – performance
 - Non-functional – design
 - Non-functional – commercial, legal(6 marks)
5. Identify which of the following statements are functional or non-functional requirement:
 - a) The software automatically validates customers against the ABC Contact Management System.
 - b) Users must change the initially assigned login password immediately after the first successful login. Moreover, the initial should never be reused.
 - c) The Sales system should allow users to record customers sales
 - d) Employees never allowed to update their salary information. Such attempt should be reported to the security administrator.
 - e) The background color for all windows in the application will be blue and have a hexadecimal RGB color value of 0x0000FF.
 - f) Every unsuccessful attempt by a user to access an item of data shall be recorded on an audit trail.
 - g) The software system should be integrated with banking API
 - h) Only Managerial level employees have the right to view revenue data.
 - i) The software should be portable. So, moving from one OS to other OS does not create any problem.
 - j) Privacy of information, the export of restricted technologies, intellectual property rights, etc. should be audited.(10 marks)

6. The customers of Just a Line are required to fill up an order form (Figure 1) in order to place the items to order from the company. Write the data dictionary notations describing the form.

Just a Line						
ORDER FORM						
Mr/Mrs/Miss/Ms _____						
Address _____						
Delivery Address _____						
Date Required _____ Telephone _____						
Description	Packet Size	No. Packets	Price		£	p
Total Cost						
Delivery Charge						
Amount Owning						

Figure 1: Form

(11 marks)

7. Explain the data elements in data dictionary.

(8 marks)

SECTION B

[50 MARKS]

Case Study: E-Commerce Website for Beautiful Bags

You are developing an e-commerce website for Beautiful Bags, a small business that sells women bags and wallets. Beautiful Bags has many varieties of collection in brands, designs, and patterns; and the business owner would like to feature all that it has.

The business owner wants a simple system where the main features are the catalogue and the shopping cart:

The catalogue:

The customer can view all of the items from the catalogue. The catalogue offers the customer an option to search for a specific brand, design or pattern. When viewing an item, the customer can click on the "buy" button to make an order.

The shopping cart:

The customer can view all of the ordered items in the shopping cart. The shopping cart offers the customer an option to modify the order and an option to make the payment.

According to the business owner, when a customer makes an order, the system should first check the stock to confirm the item's availability before adding the order in the shopping cart. A notification will be displayed should the item is not available. But if the ordered item is available, then the system will reserve the item, and add the order to the shopping cart. The reservation will last for 24 hours or upon payment. Free delivery service is applicable for each single purchase of RM300.00 or more.

1. Draw a use case diagram for e-commerce website for Beautiful Bags. (14 marks)
2. Draw a generic sequence diagram illustrating the sequence of ordering an item. (11 marks)
3. Draw an activity diagram when a customer is online-shopping with a swim lane between the user, the system, and the database. The term online-shopping is in this context includes making an order and making a payment. (15 marks)
4. Draw a class diagram that supports the description for question (2), showing the following notation between classes: association, aggregation, composition and generalization. (10 marks)

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